Your house: light, sun, heat and wind

Simulation report for the house in Maringá, PR (lat -23.41, lon -51.93) · 15 October 2026
Weather: typical year from BRA_PR_Maringa.868990_TMYx.2011-2025.epw · annual sun 1,882 kWh/m² · mean temperature 23.0 °C · mean wind 2.3 m/s

How to read this report. Every number comes with a coloured tag telling you whether it is good or bad, on a **level from 1 (bad) to 5 (excellent)**:

Excellent 5/5 Good 4/5 Acceptable 3/5 Poor 2/5 Bad 1/5

Each tag is based on a **Brazilian norm (ABNT)** where one exists, on an **international standard** where Brazil gives no number, or on a clearly labelled **rule of thumb**. The boxes *What counts as good?* at the end of each section show the full scale and the source. The model walls are assumed to be: *Ceramic hollow brick 8-hole 9x19x19 cm + 2 cm plaster both sides = 13 cm wall (standard Brazilian block)*; roof: *Clay tile + ventilated attic + 12 cm concrete slab ceiling*. These materials and the window sizes (common catalogue sizes of about 2012) are estimates, not measurements: use the results to compare rooms and options, not as a guarantee.

1. The short version

Each room is rated on up to nine checks, each on a **level from 1 (bad) to 5 (excellent)**; the OVERALL column is the average level. The next pages explain every check; the cooling column shows what an air conditioner would cost per year:

Room	Daylight (NBR 15575)	Lamps off in daytime	Sun on window	Comfort hours	Hot day	Cold day	Openable window	Cooling cost	OVERALL
Living + dining	Superior 5/5	Almost always 5/5	Below minimum 2/5	Good 4/5	Intermediate 4/5	Superior 5/5	Fails 1/5	R$ 838/yr Low 4/5	Good 3.8/5
Hallway	No requirement 3/5	Almost never 1/5	-	Excellent 5/5	-	-	-	-	Acceptable 3.0/5
Bedroom 1	Superior 5/5	Almost always 5/5	High 5/5	Good 4/5	Intermediate 4/5	Superior 5/5	Fails 1/5	R$ 169/yr Low 4/5	Good 4.0/5
Bathroom 1	Superior 5/5	Almost never 1/5	-	Acceptable 3/5	-	-	-	-	Acceptable 3.0/5
Bedroom 2	Intermediate 4/5	Often 4/5	Below minimum 2/5	Good 4/5	Superior 5/5	Superior 5/5	Fails 1/5	R$ 116/yr Very low 5/5	Good 3.7/5
Bathroom 2	Intermediate 4/5	Almost never 1/5	-	Good 4/5	-	-	-	-	Acceptable 3.0/5
Bedroom 3	Intermediate 4/5	Often 4/5	Below minimum 2/5	Good 4/5	Superior 5/5	Superior 5/5	Fails 1/5	R$ 112/yr Very low 5/5	Good 3.7/5
Kitchen	Superior 5/5	Almost always 5/5	Below minimum 2/5	Acceptable 3/5	-	-	Meets minimum 3/5	R$ 365/yr Moderate 3/5	Good 3.7/5
Outdoor bath	-	Almost always 5/5	-	Acceptable 3/5	-	-	-	-	Good 4.0/5
Outdoor storage	-	Almost always 5/5	-	Acceptable 3/5	-	-	-	-	Good 4.0/5
Whole house								**R$ 1,600/yr**	Good 3.7/5

What works well

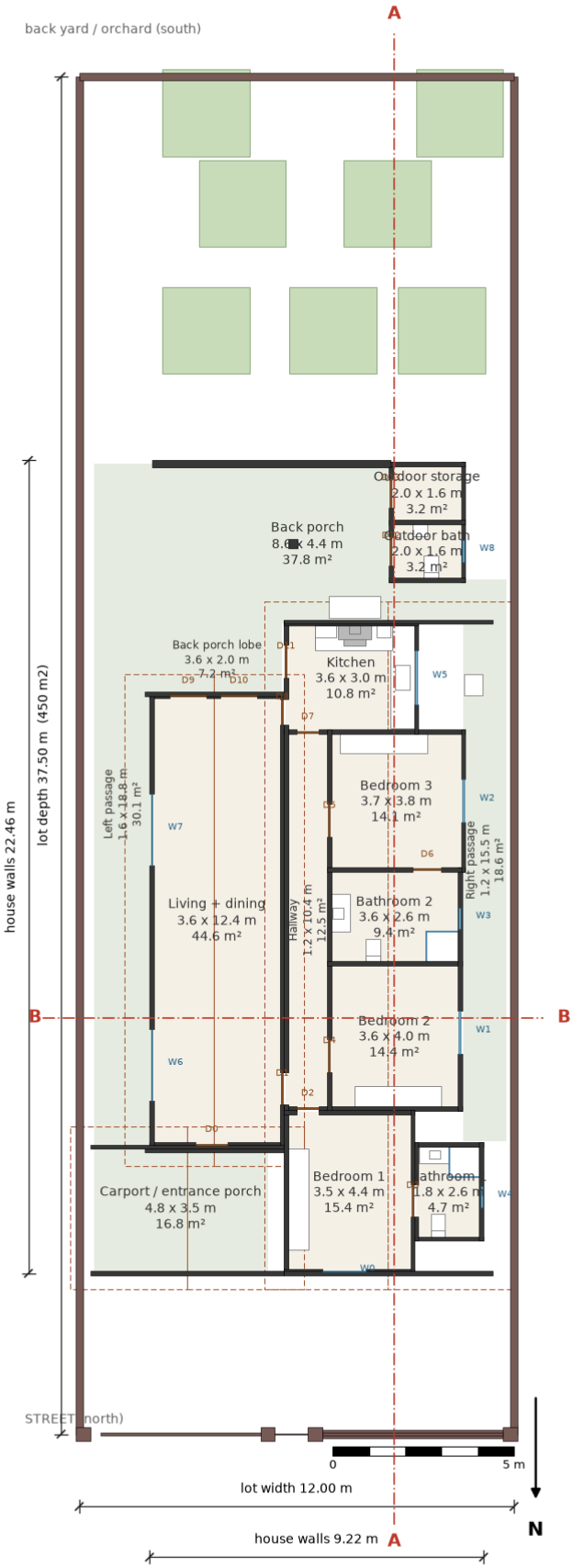
- Living (front half): 290 lux of daylight at the centre (NBR 15575 superior level)
- TV / dining (rear half): 301 lux of daylight at the centre (NBR 15575 superior level)
- Bedroom 1: 657 lux of daylight at the centre (NBR 15575 superior level)
- Kitchen: 232 lux of daylight at the centre (NBR 15575 superior level)
- Most comfortable room: Hallway (92% of the year in the comfort band; outdoors only 48%)
- External wall heat transmittance U: 2.52 W/m²K meets the limit <= 3.7
- External wall heat capacity: 134.0 kJ/m²K meets the limit >= 130

What deserves attention

- Least comfortable room: Outdoor bath (68%; 1944 h too hot, 863 h too cold)
- Living + dining: openable window area only 5.4% of the floor (NBR 15575-4 asks 7%)
- Bedroom 1: openable window area only 4.7% of the floor (NBR 15575-4 asks 7%)
- Bedroom 2: openable window area only 5.0% of the floor (NBR 15575-4 asks 7%)
- Bedroom 3: openable window area only 5.1% of the floor (NBR 15575-4 asks 7%)
- Roof heat transmittance U (indicative): 1.78 W/m²K vs limit <= 1.5

FLOOR PLAN - ground floor

House faces NORTH (street at the bottom) - rooms with name, size and area; sections A-A a



Scale 1:150 - sizes are estimates from a video scan and the listing, not a survey

Floor plan of the model (north / street at the bottom).

2. Natural light

Daylight saves electricity and makes rooms pleasant. The Brazilian norm NBR 15575 asks for at least **60 lux** at the centre of living rooms, bedrooms and kitchens (lux = how much light lands on a table). For comparison: a bright overcast day outdoors is thousands of lux.

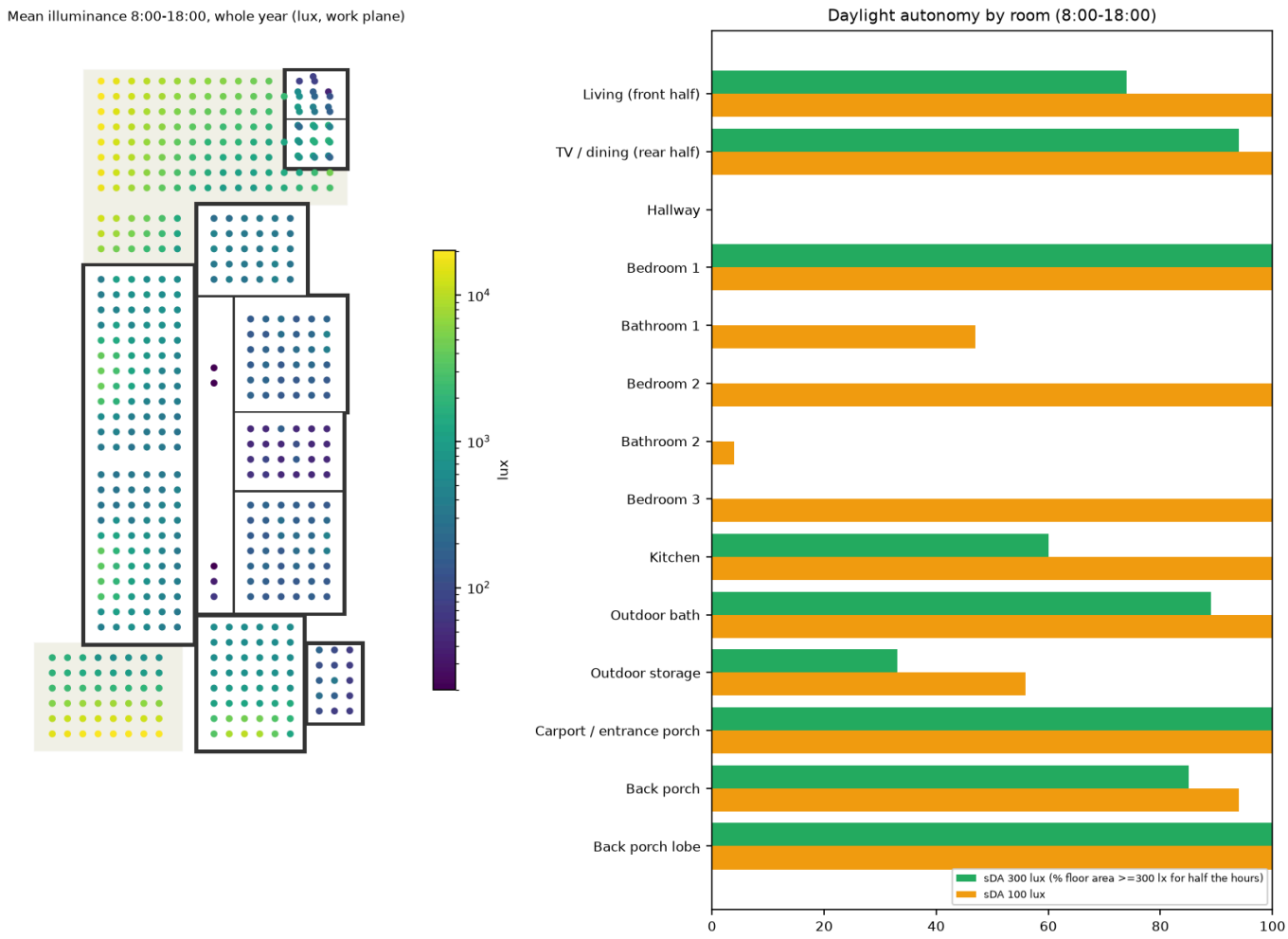
2.1 Brazilian test: light at the centre of the room, lamps off

Room	Lux at centre	Verdict
Living (front half)	290	Superior 5/5
TV / dining (rear half)	301	Superior 5/5
Hallway	0	No requirement 3/5
Bedroom 1	657	Superior 5/5
Bathroom 1	63	Superior 5/5
Bedroom 2	97	Intermediate 4/5
Bathroom 2	32	Intermediate 4/5
Bedroom 3	115	Intermediate 4/5
Kitchen	232	Superior 5/5
Outdoor bath	262	Superior 5/5
Outdoor storage	90	Intermediate 4/5

2.2 How much of the room is bright, and when do I need a lamp?

Room	Goal (lux)	Floor at the goal	Daytime with lamps off	Strong sun (ASE)
Living (front half)	150	100% Whole room 5/5	93% Almost always 5/5	5% Acceptable 4/5
TV / dining (rear half)	150	100% Whole room 5/5	95% Almost always 5/5	7% Acceptable 4/5
Hallway	-	-	0% Almost never 1/5	0% Very low 5/5
Bedroom 1	100	100% Whole room 5/5	98% Almost always 5/5	21% Very high 1/5
Bathroom 1	-	-	8% Almost never 1/5	0% Very low 5/5
Bedroom 2	100	100% Whole room 5/5	58% Often 4/5	0% Very low 5/5
Bathroom 2	-	-	2% Almost never 1/5	0% Very low 5/5
Bedroom 3	100	100% Whole room 5/5	60% Often 4/5	0% Very low 5/5
Kitchen	200	100% Whole room 5/5	88% Almost always 5/5	0% Very low 5/5
Outdoor bath	-	-	95% Almost always 5/5	0% Very low 5/5
Outdoor storage	-	-	87% Almost always 5/5	0% Very low 5/5

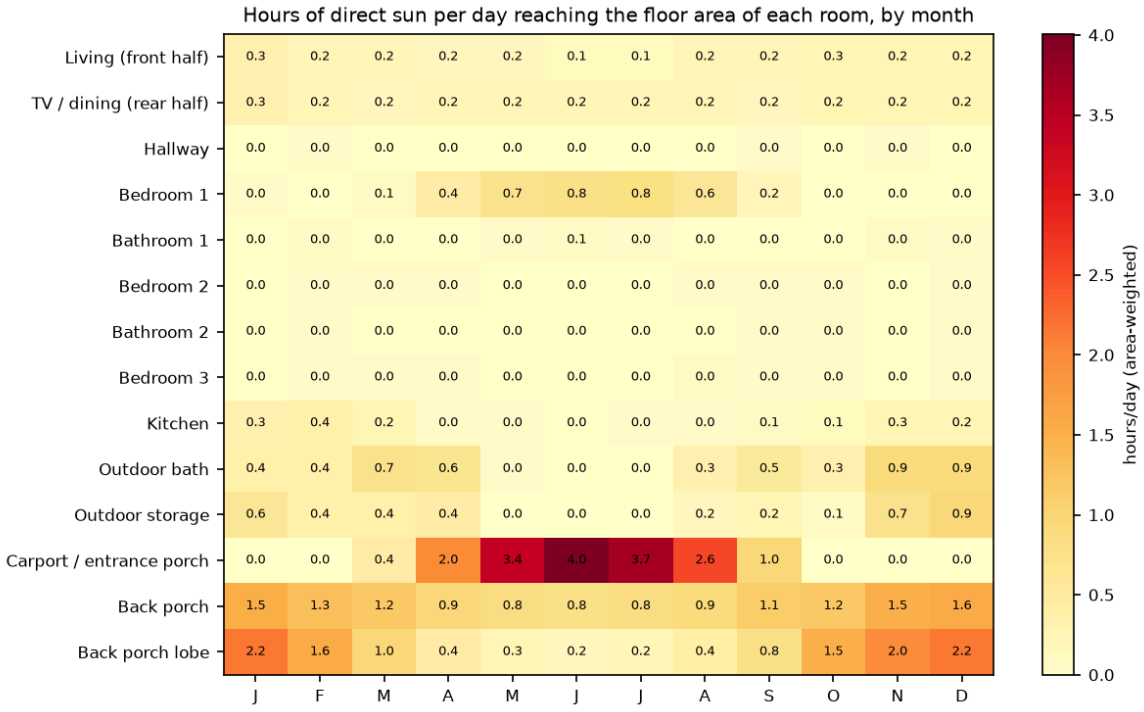
Goal = the light level EN 17037 (UK National Annex for dwellings) recommends from daylight: 100 lux bedroom, 150 lux living room, 200 lux kitchen. 'Lamps off' = share of the hours 8:00-18:00 in which the daylight average already reaches the general light level of section 3.



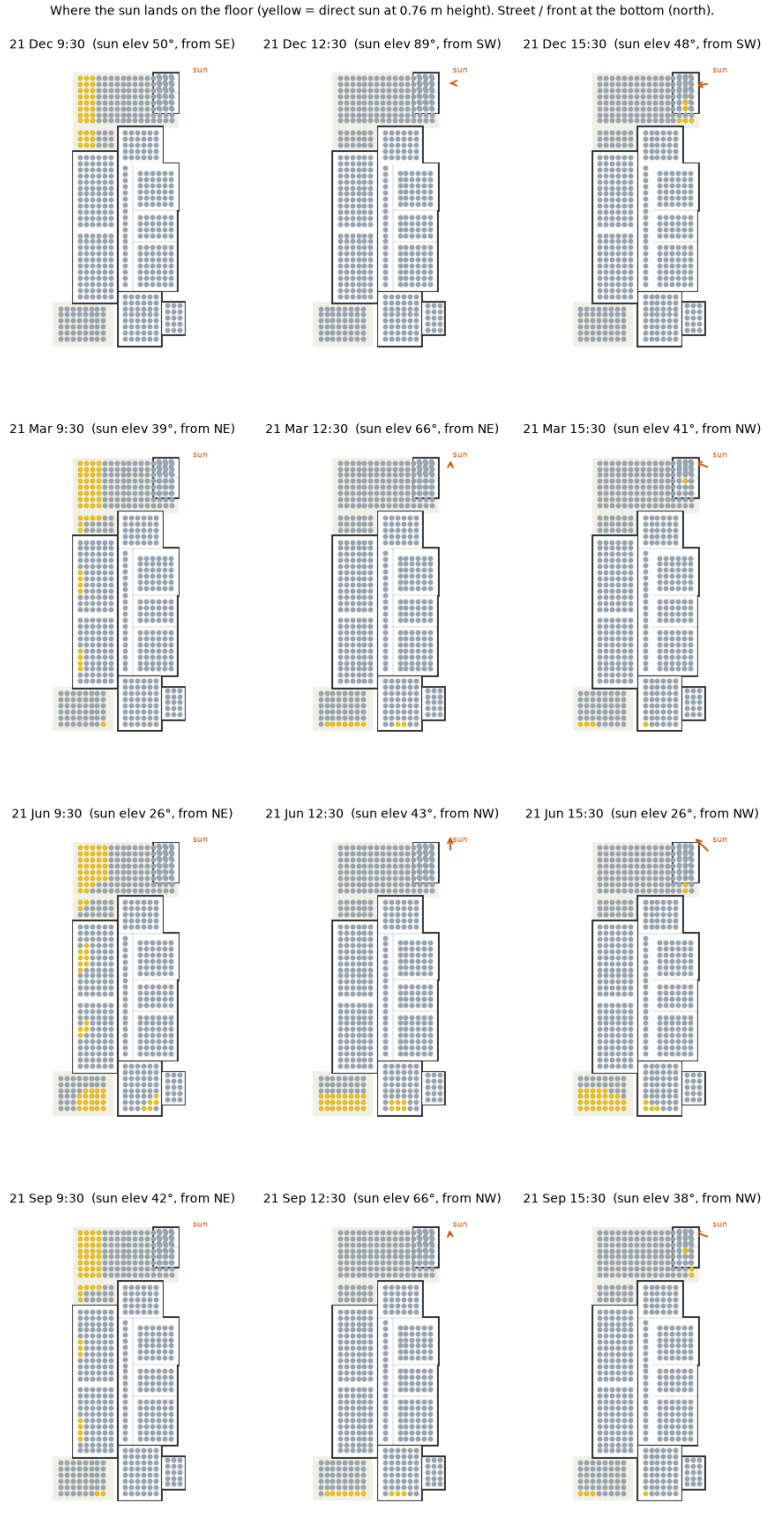
Left: average light on the floor over the year. Right: share of each room that reaches 100 and 300 lux.

2.3 Sun reaching the windows

Room	Sun on window, hours/day (Aug-Sep)	Verdict
Living (front half)	0.5	Below minimum 2/5
TV / dining (rear half)	0.9	Below minimum 2/5
Bedroom 1	10.7	High 5/5
Bedroom 2	0.0	Below minimum 2/5
Bedroom 3	0.0	Below minimum 2/5
Kitchen	0.0	Below minimum 2/5



Hours per day of direct sun that reach the floor of each room, by month (J = January).



Where the sun lands on the floor on four key dates.

What counts as good? (light)

Daylight at the centre of the room (NBR 15575 test) Brazilian norm

Natural light on a table-height plane (0.75 m) at the middle of the room, on 23 April and 23 October at 9:30 and 15:30 with a half-cloudy sky, lamps off. The lowest of the four readings counts. Applies to living rooms, bedrooms, kitchens and service areas.

Fails 1/5	below 60 lux	Below the 60 lux minimum: you need a lamp even at midday.
Minimum 3/5	60 to 90 lux	Meets the legal minimum of the norm.
Intermediate 4/5	90 to 120 lux	Better than the minimum.
Superior 5/5	120 or more lux	Top level of the norm.

This house: Living (front half) 290 lux Superior 5/5 | TV / dining (rear half) 301 lux Superior 5/5 | Bedroom 1 657 lux Superior 5/5 | Bedroom 2 97 lux Intermediate 4/5 | Bedroom 3 115 lux Intermediate 4/5 | Kitchen 232 lux Superior 5/5

Source: ABNT NBR 15575-1:2013 (Residential buildings - Performance), daylight and thermal requirements. Our run uses the typical weather of the week around those dates and sky light only (no direct sun), so it is on the safe side.

Daylight at the centre of the room (NBR 15575 test) - bathrooms, hallways, stairs, garages Brazilian norm

Same test as above, but for rooms where people do not stay long.

No requirement 3/5	below 30 lux	The norm does not require daylight here (minimum level: 'not required').
Intermediate 4/5	30 to 45 lux	Meets the intermediate level.
Superior 5/5	45 or more lux	Meets the superior level.

This house: Hallway 0 lux No requirement 3/5 | Bathroom 1 63 lux Superior 5/5 | Bathroom 2 32 lux Intermediate 4/5

Source: ABNT NBR 15575-1:2013 (Residential buildings - Performance), daylight and thermal requirements.

Share of the floor that reaches the home target light level International standard

Percentage of the room's floor that gets at least the target light (100 lux bedroom, 150 lux living room, 200 lux kitchen) from daylight alone, for at least half of the daytime hours (8:00-18:00).

Below target 2/5	below 50 % of floor	Less than half of the room reaches the target: lamps will be needed in the daytime.
Meets target 4/5	50 to 95 % of floor	At least half of the floor reaches the target for half of the day.
Whole room 5/5	95 or more % of floor	Practically the entire room reaches the target.

This house: Living (front half) 100% Whole room 5/5 | TV / dining (rear half) 100% Whole room 5/5 | Bedroom 1 100% Whole room 5/5 | Bedroom 2 100% Whole room 5/5 | Bedroom 3 100% Whole room 5/5 | Kitchen 100% Whole room 5/5

Source: EN 17037:2018 Daylight in buildings (UK National Annex dwelling targets 100/150/200 lx; sunlight 1.5/3/4 h). EN 17037 asks for the target over at least 50% of the area for half of the daylight hours (+ a lower level over 95%).

Spatial daylight autonomy sDA 300/50% (work-space standard) International standard

Percentage of the floor with at least 300 lux of daylight for at least half of the occupied hours. 300 lux is a WORKING-light level, so this is a strict check: useful for a home office, kitchen or study; bedrooms and hallways are not expected to reach it.

Below recommendation 2/5	below 55 % of floor	Less than 55% of the floor reaches 300 lux for half of the day.
Nominally acceptable 3/5	55 to 75 % of floor	Accepted by the IES / LEED (55% or more).
Preferred 5/5	75 or more % of floor	Preferred level (75% or more).

This house: Living (front half) 74% Nominally acceptable 3/5 | TV / dining (rear half) 94% Preferred 5/5 | Hallway 0% Below recommendation 2/5 | Bedroom 1 100% Preferred 5/5 | Bathroom 1 0% Below recommendation 2/5 | Bedroom 2 0% Below recommendation 2/5 | Bathroom 2 0% Below recommendation 2/5 | Bedroom 3 0% Below recommendation 2/5 | Kitchen 60% Nominally acceptable 3/5 | Outdoor bath 89% Preferred 5/5 | Outdoor storage 33% Below recommendation 2/5

Source: IES LM-83-12 (sDA and ASE), as used by LEED v4.1.

Annual sunlight exposure ASE (glare and overheating risk) International standard

Percentage of the floor that receives strong direct sun (over 1000 lux) for more than 250 hours a year. High values mean glare, faded furniture and heat.

Very low 5/5	below 3 % of floor	Practically no harsh direct sun.
Acceptable 4/5	3 to 10 % of floor	Within the 10% limit used by IES / LEED.
High 2/5	10 to 20 % of floor	Above the 10% limit: consider curtains, blinds or an eave.
Very high 1/5	20 or more % of floor	Large area with long strong sun: glare and heat likely.

This house: Living (front half) 5% Acceptable 4/5 | TV / dining (rear half) 7% Acceptable 4/5 | Hallway 0% Very low 5/5 | Bedroom 1 21% Very high 1/5 | Bathroom 1 0% Very low 5/5 | Bedroom 2 0% Very low 5/5 | Bathroom 2 0% Very low 5/5 | Bedroom 3 0% Very low 5/5 | Kitchen 0% Very low 5/5 | Outdoor bath 0% Very low 5/5 | Outdoor storage 0% Very low 5/5

Source: IES LM-83-12 (sDA and ASE), as used by LEED v4.1.

Daytime hours without needing lamps Rule of thumb

Share of the daytime hours in which the room's average daylight already reaches the recommended general light level (see the lamp table), so the lights can stay off.

Almost never 1/5	below 10 % of 8:00-18:00 hours	You will switch the lights on all day.
Rarely 2/5	10 to 25 % of 8:00-18:00 hours	Lamps needed most of the day.
Sometimes 3/5	25 to 50 % of 8:00-18:00 hours	Lamps needed on dull days and mornings / evenings.
Often 4/5	50 to 75 % of 8:00-18:00 hours	Lamps mostly off during the day.
Almost always 5/5	75 or more % of 8:00-18:00 hours	Lamps off nearly the whole day.

This house: Living (front half) 93% Almost always 5/5 | TV / dining (rear half) 95% Almost always 5/5 | Hallway 0% Almost never 1/5 | Bedroom 1 98% Almost always 5/5 | Bathroom 1 8% Almost never 1/5 | Bedroom 2 58% Often 4/5 | Bathroom 2 2% Almost never 1/5 | Bedroom 3 60% Often 4/5 | Kitchen 88% Almost always 5/5 | Outdoor bath 95% Almost always 5/5 | Outdoor storage 87% Almost always 5/5

Source: Rule of thumb of this study (no norm gives a number); ABNT NBR 5413:1992 (withdrawn in 2013 but still the usual Brazilian reference for HOME lighting levels).

Direct sun on the window in early spring (August-September here) International standard

Hours of direct sun that reach the room's windows on a typical day between 1 August and 21 September (EN 17037 uses 1 Feb-21 Mar in the northern hemisphere). Sun in winter / early spring is welcome for health and warmth.

Below minimum 2/5	below 1.5 hours/day	Less than 1.5 h of sun: the room feels sunless.
Minimum 3/5	1.5 to 3 hours/day	Meets the EN 17037 minimum (1.5 h).
Medium 4/5	3 to 4 hours/day	Meets the EN 17037 medium level (3 h).
High 5/5	4 or more hours/day	Meets the EN 17037 high level (4 h).

This house: Living (front half) 0.5 h Below minimum 2/5 | TV / dining (rear half) 0.9 h Below minimum 2/5 | Bedroom 1 10.7 h High 5/5 | Bedroom 2 0.0 h Below minimum 2/5 | Bedroom 3 0.0 h Below minimum 2/5 | Kitchen 0.0 h Below minimum 2/5

Source: EN 17037:2018 Daylight in buildings (UK National Annex dwelling targets 100/150/200 lx; sunlight 1.5/3/4 h). EN 17037 asks this for at least one main room (preferably the living room); bathrooms and hallways are exempt.

3. Lamps: how much light to install (lumens)

Lux is the light that reaches a surface; **lumens** is the light a lamp gives out. To get an average of X lux over a floor of A m² you need X × A ÷ (utilisation × maintenance = 0.45 × 0.8 = 0.36) lumens. The 'minimum' column is the bare lux × m².

3.1 Recommended light levels at home

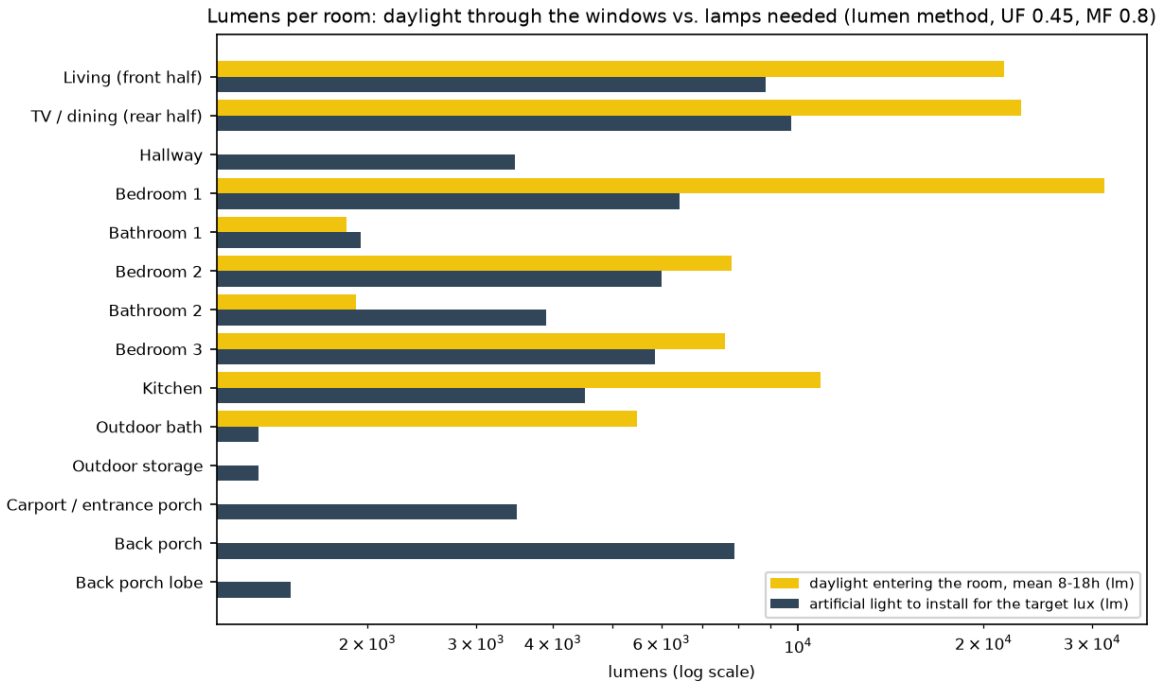
Kind of room	General light (lux): min / recommended / max	Task light (lux)	Source
Living / dining room	100 / 150 / 200	300 / 500 / 750 - reading, writing, sewing	NBR 5413 (home); EN 12464-1 / NBR 8995-1 for reading tasks
Bedroom	100 / 150 / 200	200 / 300 / 500 - reading in bed, dressing table	NBR 5413 (home)
Kitchen	100 / 150 / 200	200 / 300 / 500 - stove, sink and worktop (EN 12464-1 asks 500)	NBR 5413 (home); EN 12464-1
Bathroom	100 / 150 / 200	200 / 300 / 500 - mirror (shaving, make-up)	NBR 5413 (home)
Hallway / circulation	75 / 100 / 150	-	NBR 5413 (home)
Laundry / storage / outdoor bath	100 / 150 / 200	200 / 300 / 500 - laundry and tools	NBR 5413 (home)
Porch / carport (covered outdoor)	50 / 75 / 100	-	NBR 5413 (garages 50-75-100)

Reading: light the room to the general level (100-200 lux) and add a directional **reading lamp of about 450-800 lumens** (about 4-8 W LED) near the page to reach **300-500 lux** on it (NBR ISO/CIE 8995-1 and EN 12464-1 for reading and writing; NBR 5413 for homes). Reading under only the 100-150 lux of general light strains the eyes. For scale: 100 lm = night light / small candle-type; 450 lm = equivalent of an old 40 W bulb; 800 lm = equivalent of an old 60 W bulb; 1600 lm = equivalent of an old 100 W bulb.

3.2 Lamp lumens for each room of this house

Room	m²	General light goal	Task light (reading etc.)	Lamp lumens: minimum - design	LED watts (design)
Living (front half)	21.2	150 lux	500 lux (reading, writing, sewing)	3,186 - 8,850	88
TV / dining (rear half)	23.4	150 lux	500 lux (reading, writing, sewing)	3,510 - 9,750	98
Hallway	12.5	100 lux	-	1,248 - 3,467	35
Bedroom 1	15.4	150 lux	300 lux (reading in bed, dressing table)	2,310 - 6,417	64
Bathroom 1	4.7	150 lux	300 lux (mirror (shaving, make-up))	702 - 1,950	19
Bedroom 2	14.4	150 lux	300 lux (reading in bed, dressing table)	2,160 - 6,000	60
Bathroom 2	9.4	150 lux	300 lux (mirror (shaving, make-up))	1,404 - 3,900	39
Bedroom 3	14.1	150 lux	300 lux (reading in bed, dressing table)	2,109 - 5,858	59
Kitchen	10.8	150 lux	300 lux (stove, sink and worktop (EN 12464-1 asks 500))	1,620 - 4,500	45
Outdoor bath	3.2	150 lux	300 lux (laundry and tools)	480 - 1,333	13
Outdoor storage	3.2	150 lux	300 lux (laundry and tools)	480 - 1,333	13

Use the low end for light walls and ceilings and good fittings; the high end for dark colours or recessed spots. LED at 100 lm per watt. Daylight already covers part of the day (section 2.2).



Daylight entering through the windows (yellow) compared with the lamp lumens needed for the target light (dark).

4. Windows, fresh air and wind

NBR 15575-4 asks that the openable part of the windows is at least **7% of the floor area** of each living room and bedroom, and limits glass to 20% of the floor unless it is shaded. NBR 15220-3 recommends wind-driven cross-ventilation as the main summer strategy for this climate.

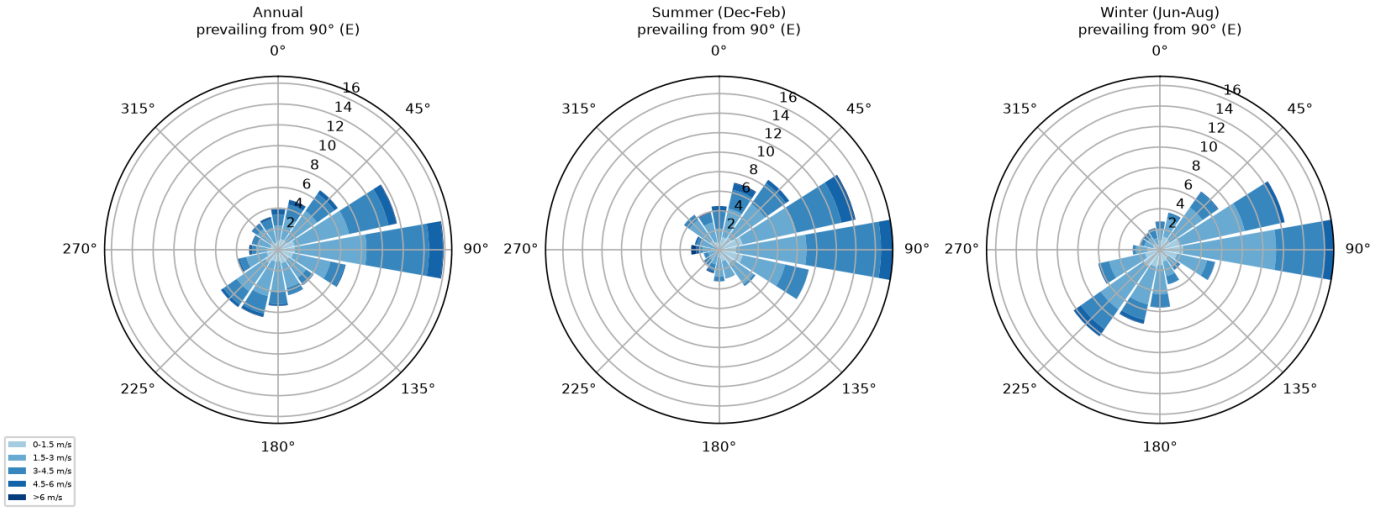
4.1 Window size

Room	Floor m²	Glass m²	Glass / floor	Openable / floor
Living + dining	44.6	4.80	10.8% Small 3/5	5.4% Fails 1/5
Hallway	12.5	0.00	0.0%	0.0%
Bedroom 1	15.4	1.44	9.4% Small 3/5	4.7% Fails 1/5
Bathroom 1	4.7	0.36	7.7%	3.8%
Bedroom 2	14.4	1.44	10.0% Small 3/5	5.0% Fails 1/5
Bathroom 2	9.4	0.36	3.8%	1.9%
Bedroom 3	14.1	1.44	10.2% Small 3/5	5.1% Fails 1/5
Kitchen	10.8	1.80	16.7% Ideal 5/5	8.3% Meets minimum 3/5
Outdoor bath	3.2	0.36	11.2%	5.6%
Outdoor storage	3.2	0.00	0.0%	0.0%

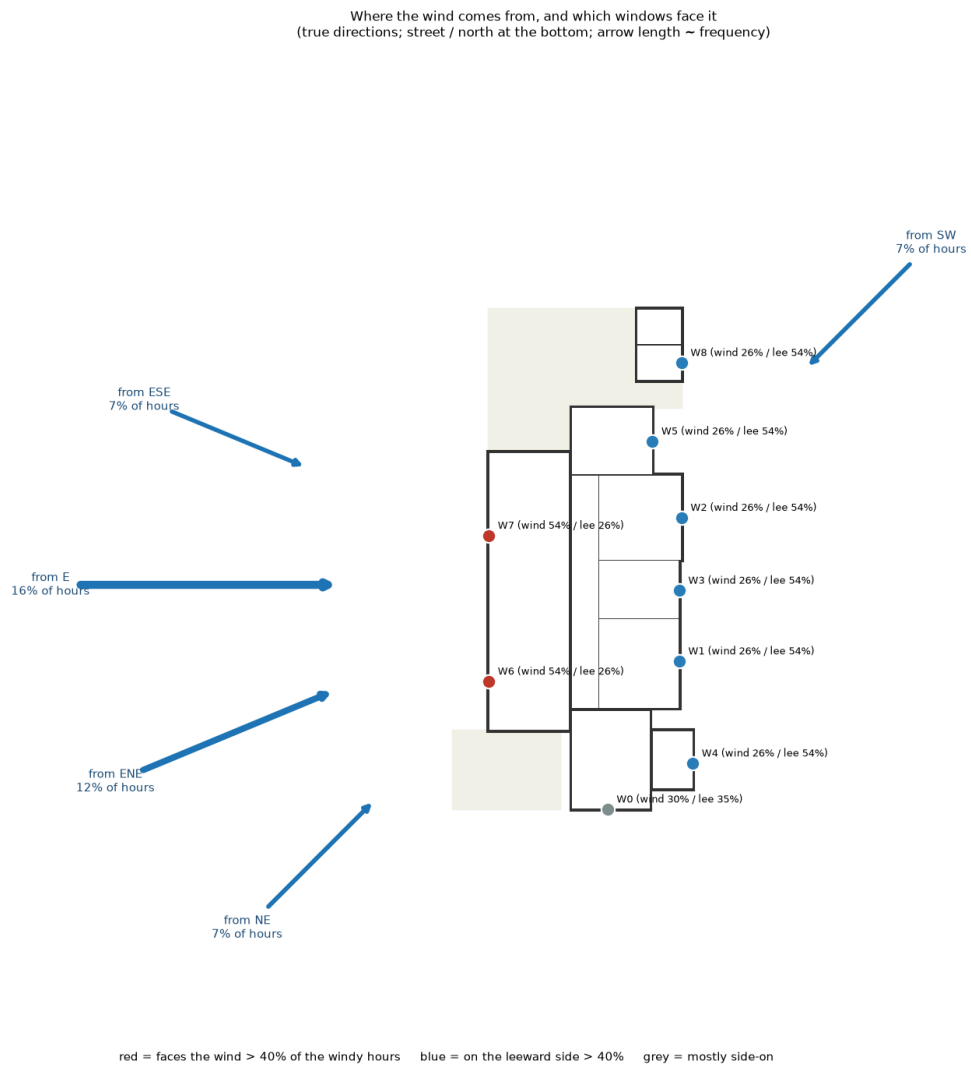
Window sizes in the model are common catalogue sizes of about 2012 (bedrooms 1.20 x 1.20 m, living and dining 2.00 x 1.20 m, kitchen 1.50 x 1.20 m, bathrooms 0.60 x 0.60 m), not measurements; a sliding window is assumed to open half of its area. If the real windows differ, change them and rerun the analysis.

4.2 Cross-ventilation

Room	Windy hours with a through-draught
Living + dining	80% Excellent 5/5
Hallway	80%
Bedroom 1	80% Excellent 5/5
Bathroom 1	29%
Bedroom 2	80% Excellent 5/5
Bathroom 2	0%
Bedroom 3	80% Excellent 5/5
Kitchen	80% Excellent 5/5
Outdoor bath	0%
Outdoor storage	0%



Wind roses: where the wind comes from and how strong, whole year, summer and winter.



Wind directions against the windows of the house (red windows face the wind, blue are on the sheltered side).

What counts as good? (windows and wind)

Openable window area for ventilation Brazilian norm

Area of windows that can really open (glass, frames and fixed leaves are discounted), divided by the floor area of the room. Zones 1-7 need at least 7%.

Fails 1/5	below 7 % of floor area	Less than the 7% minimum of the norm (unless the city's building code says otherwise).
Meets minimum 3/5	7 to 10 % of floor area	Meets the 7% minimum.
Comfortable margin 5/5	10 or more % of floor area	More than the minimum: good air changes.

This house: Living + dining 5.4% Fails 1/5 | Bedroom 1 4.7% Fails 1/5 | Bedroom 2 5.0% Fails 1/5 | Bedroom 3 5.1% Fails 1/5 | Kitchen 8.3% Meets minimum 3/5

Source: ABNT NBR 15575-4:2021 amendment draft (Mar 2021): simplified method for walls, ventilation, transparent areas. We assume a sliding window opens about half of its area; a casement / tilting window opens more.

Glass area compared with the floor Brazilian norm

Window area divided by floor area. NBR 15575-4 limits transparent areas to 20% of the floor (rooms up to 20 m2) unless there is shading or high-performance glass; NBR 15220-3 recommends 15-25% openings for zone 3.

Very small 2/5	below 7 % of floor area	Too little window for light and air.
Small 3/5	7 to 15 % of floor area	Works, but a little dark or stuffy.
Ideal 5/5	15 to 20 % of floor area	Inside the NBR 15575-4 limit and the NBR 15220-3 recommendation.
Large 4/5	20 to 25 % of floor area	Recommended by NBR 15220-3 if shaded; above the NBR 15575-4 20% unless shaded.
Too large 2/5	25 or more % of floor area	More sun heat than the norms accept without shading.

This house: Living + dining 10.8% Small 3/5 | Bedroom 1 9.4% Small 3/5 | Bedroom 2 10.0% Small 3/5 | Bedroom 3 10.2% Small 3/5 | Kitchen 16.7% Ideal 5/5

Source: ABNT NBR 15575-4:2021 amendment draft (Mar 2021): simplified method for walls, ventilation, transparent areas; ABNT NBR 15220-3: Brazilian bioclimatic zoning and building guidelines (Maringa is generally listed in zone 3).

Cross-ventilation possible Rule of thumb

Share of the windy hours in which the wind hits a window on one side of the connected rooms and another window on the opposite side can let it out (doors assumed open).

Almost none 1/5	below 5 % of windy hours	Air hardly flows through the room.
Weak 2/5	5 to 20 % of windy hours	Cross-ventilation seldom happens.
Some 3/5	20 to 40 % of windy hours	Works in part of the windy hours.
Good 4/5	40 to 60 % of windy hours	Works in many windy hours.
Excellent 5/5	60 or more % of windy hours	A through-draught is possible most of the time.

This house: Living + dining 80% Excellent 5/5 | Bedroom 1 80% Excellent 5/5 | Bedroom 2 80% Excellent 5/5 | Bedroom 3 80% Excellent 5/5 | Kitchen 80% Excellent 5/5

Source: Rule of thumb of this study (no norm gives a number).

Average outdoor wind speed International standard

Average wind at the site over the year (weather file, open terrain at 10 m; at the house it is lower).

Calm 2/5	below 1.0 m/s	Little natural ventilation from the wind alone.
Light air 3/5	1.0 to 2.0 m/s	Gentle; openings help but cross-ventilation is weak.
Light to gentle breeze 5/5	2.0 to 5.0 m/s	Ideal range for natural ventilation.
Moderate breeze 4/5	5.0 to 8.0 m/s	Strong enough to need wind protection outdoors.
Fresh breeze or more 2/5	8.0 or more m/s	Wind damage and discomfort risk.

This house: Maringá, whole year 2.3 m/s Light to gentle breeze 5/5

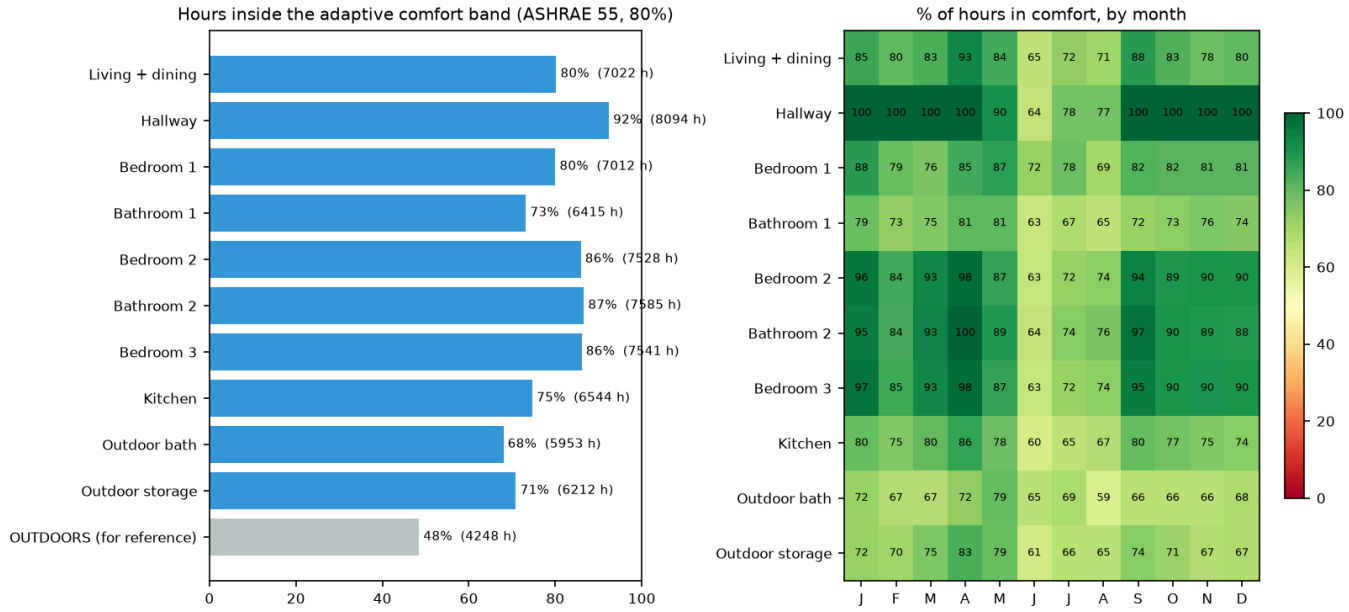
Source: WMO Beaufort wind scale; Rule of thumb of this study (no norm gives a number). Beaufort 1 = 0.3-1.5 m/s, 2 = 1.6-3.3, 3 = 3.4-5.4, 4 = 5.5-7.9 m/s. Indoors ASHRAE-55 accepts air speeds up to about 0.8 m/s.

5. Temperature and comfort

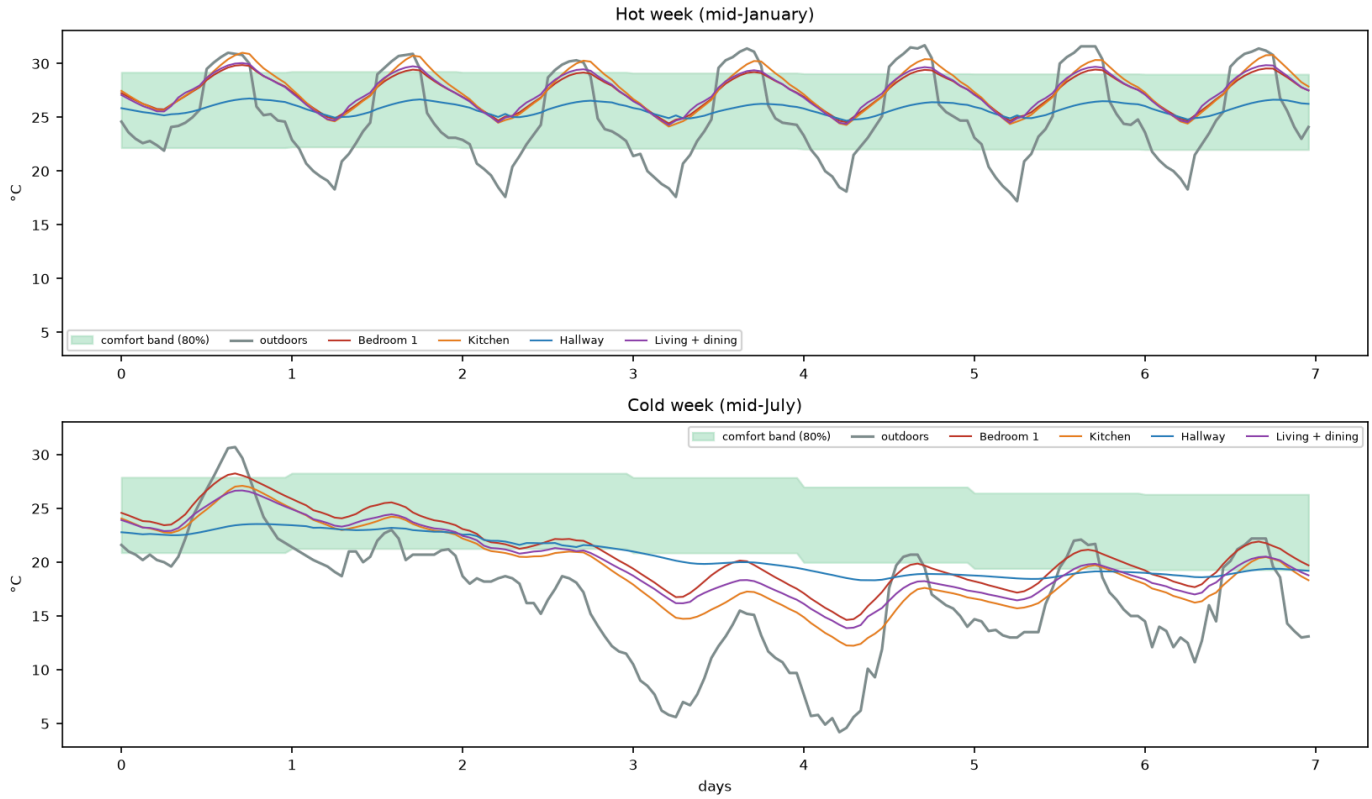
Without air conditioning, comfort depends on the walls, roof, windows, shade and ventilation. The simulation computes the indoor temperature of every room for each of the 8,760 hours of a typical year. **Comfortable** means inside the ASHRAE-55 adaptive comfort band (it moves with the weather: people accept warmer rooms when it is warmer outside).

5.1 Hours in the comfortable band

Room	Comfortable hours	Too hot	Too cold	Hottest	Coldest
Living + dining	7,022 h = 80% Good 4/5	885 h	853 h	33.7 °C Very hot 2/5	12.9 °C Cold 2/5
Hallway	8,094 h = 92% Excellent 5/5	0 h	666 h	29.1 °C Warm 4/5	16.5 °C Cool 3/5
Bedroom 1	7,012 h = 80% Good 4/5	1,102 h	646 h	33.6 °C Very hot 2/5	13.1 °C Cold 2/5
Bathroom 1	6,415 h = 73% Acceptable 3/5	1,376 h	969 h	35.5 °C Extreme 1/5	11.1 °C Dangerous 1/5
Bedroom 2	7,528 h = 86% Good 4/5	439 h	793 h	32.8 °C Very hot 2/5	14.0 °C Cold 2/5
Bathroom 2	7,585 h = 87% Good 4/5	463 h	712 h	32.5 °C Very hot 2/5	14.4 °C Cold 2/5
Bedroom 3	7,541 h = 86% Good 4/5	425 h	794 h	32.7 °C Very hot 2/5	14.0 °C Cold 2/5
Kitchen	6,544 h = 75% Acceptable 3/5	1,136 h	1,080 h	35.2 °C Extreme 1/5	11.5 °C Dangerous 1/5
Outdoor bath	5,953 h = 68% Acceptable 3/5	1,944 h	863 h	36.6 °C Extreme 1/5	10.9 °C Dangerous 1/5
Outdoor storage	6,212 h = 71% Acceptable 3/5	1,506 h	1,042 h	35.8 °C Extreme 1/5	10.8 °C Dangerous 1/5
OUTDOORS (for reference)	4,248 h = 48% Poor 2/5	1,193 h	3,319 h	37.7 °C	4.2 °C



Left: comfortable hours per room. Right: share of comfortable hours by month (green = good, red = bad).



A hot week and a cold week: indoor temperatures against the comfort band and the outdoor temperature.

5.2 Brazilian design-day check (NBR 15575)

Hottest day of the typical year: **02 Feb** (37.7 °C outside). Coldest day: **19 Jul** (4.2 °C outside).

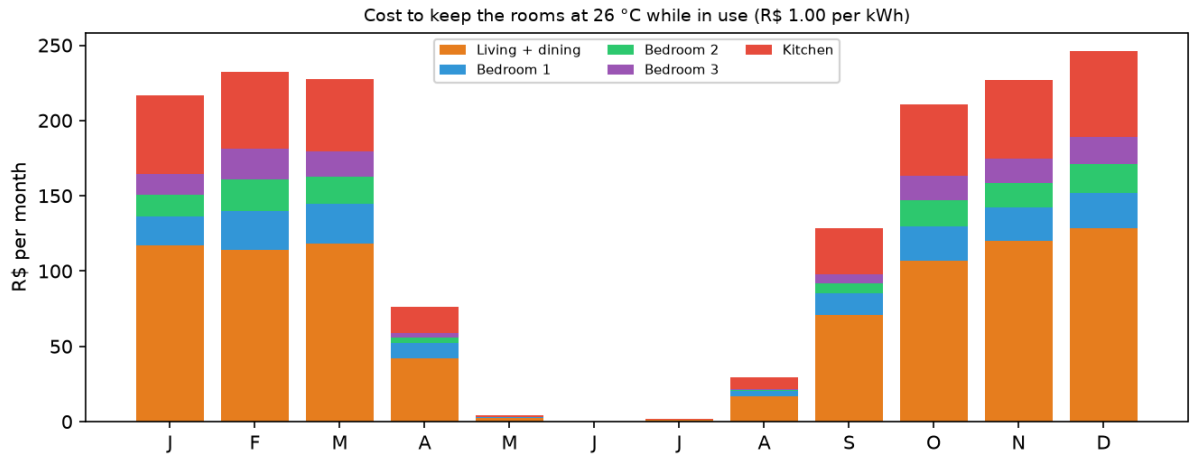
Room	Hottest day: indoor peak	vs. outdoors	Verdict	Coldest day: indoor minimum	vs. outdoors	Verdict
Living + dining	33.8 °C	-3.9	Intermediate 4/5	13.6 °C	+9.4	Superior 5/5
Hallway *(not rated)*	29.1 °C	-8.6		18.3 °C	+14.1	
Bedroom 1	33.7 °C	-4.0	Intermediate 4/5	14.5 °C	+10.3	Superior 5/5
Bathroom 1 *(not rated)*	35.7 °C	-2.0		11.9 °C	+7.7	
Bedroom 2	32.9 °C	-4.8	Superior 5/5	14.9 °C	+10.7	Superior 5/5
Bathroom 2 *(not rated)*	32.5 °C	-5.2		15.6 °C	+11.4	
Bedroom 3	32.8 °C	-4.9	Superior 5/5	15.0 °C	+10.8	Superior 5/5
Kitchen *(not rated)*	35.4 °C	-2.3		12.1 °C	+7.9	
Outdoor bath *(not rated)*	36.7 °C	-1.0		11.9 °C	+7.7	
Outdoor storage *(not rated)*	35.8 °C	-1.9		11.5 °C	+7.3	

The newest edition (NBR 15575-1:2021) compares the house with a reference house over the whole year; that needs a second simulation, so this report uses the classic absolute design-day criteria, which are easier to understand.

5.3 What it costs to cool the rooms in summer

If a room is too hot you can cool it with an air conditioner. The simulation switches on a split air conditioner whenever the room is above **26 °C** during the hours it is used (bedroom: 21:00-08:00, living: 14:00-23:00, kitchen: 08:00-21:00), with windows closed, and adds up the electricity: heat removed ÷ COP 3.0 (an inverter model does better) × tariff **R$ 1.00 per kWh** (all taxes in; an assumption based on the Copel residential tariff after the +20% adjustment of June 2026; change it with *run_all.bat --tariff 0.95*).

Room	When it is cooled	Hours the A/C runs	Electricity kWh/year	Cost R$/year	Cost Dec-Feb	Worst month	Peak heat to remove	A/C size	Level
Living + dining (44.6 m²)	14:00-23:00	1634	838	**R$ 838**	R$ 360	Dec (R$ 129)	5.6 kW	24,000 BTU/h	Low 4/5
Bedroom 1 (15.4 m²)	21:00-08:00	1050	169	**R$ 169**	R$ 69	Mar (R$ 27)	2.0 kW	9,000 BTU/h	Low 4/5
Bedroom 2 (14.4 m²)	21:00-08:00	919	116	**R$ 116**	R$ 54	Feb (R$ 21)	1.6 kW	9,000 BTU/h	Very low 5/5
Bedroom 3 (14.1 m²)	21:00-08:00	913	112	**R$ 112**	R$ 52	Feb (R$ 20)	1.5 kW	9,000 BTU/h	Very low 5/5
Kitchen (10.8 m²)	08:00-21:00	2173	365	**R$ 365**	R$ 160	Dec (R$ 57)	1.6 kW	9,000 BTU/h	Moderate 3/5
Total			**1,600**	**R$ 1,600**					



Monthly cost of cooling by room.

'Peak heat to remove' is the cooling power needed in the hottest hour; the A/C size adds 10% and rounds up to a catalogue size (1 kW = 3,412 BTU/h). Only sensible heat is counted (no dehumidification), so real bills are a little higher. Rooms with a High or Very high level are the ones where shade, a better roof or better insulation pay back fastest.

What counts as good? (temperature and cooling cost)

Hours in the comfortable temperature band Rule of thumb

Share of the 8,760 hours of the year in which the room temperature stays inside the ASHRAE-55 adaptive comfort band (it moves with the weather: warmer outdoors = wider acceptance). Outdoors, for reference, only about half of the hours.

Bad 1/5	below 40 % of the year	Uncomfortable most of the year.
Poor 2/5	40 to 60 % of the year	Uncomfortable much of the year.
Acceptable 3/5	60 to 75 % of the year	Comfortable most of the time, with some hot or cold spells.
Good 4/5	75 to 90 % of the year	Comfortable most of the year.
Excellent 5/5	90 or more % of the year	Comfortable nearly all year without air conditioning or heating.

This house: Living + dining 80% Good 4/5 | Hallway 92% Excellent 5/5 | Bedroom 1 80% Good 4/5 | Bathroom 1 73% Acceptable 3/5 | Bedroom 2 86% Good 4/5 | Bathroom 2 87% Good 4/5 | Bedroom 3 86% Good 4/5 | Kitchen 75% Acceptable 3/5 | Outdoor bath 68% Acceptable 3/5 | Outdoor storage 71% Acceptable 3/5 | OUTDOORS (for reference) 48% Poor 2/5

Source: ASHRAE 55 adaptive comfort model (80% / 90% acceptability); Rule of thumb of this study (no norm gives a number). The norms judge thermal comfort relative to a reference house or on design days; this percentage scale is our own.

Hottest day: indoor peak vs. outdoor peak (NBR 15575) Brazilian norm

On the hottest day of the typical year, the hottest indoor air temperature compared with the hottest outdoor temperature. Negative means the house is cooler than the outdoors.

Superior 5/5	below -4 degC (indoor max - outdoor max)	Indoor peak at least 4 degC below the outdoor peak.
Intermediate 4/5	-4 to -2 degC (indoor max - outdoor max)	Indoor peak at least 2 degC below the outdoor peak.
Minimum 3/5	-2 to 0.0001 degC (indoor max - outdoor max)	Indoor peak no higher than the outdoor peak.
Fails 1/5	0.0001 or more degC (indoor max - outdoor max)	Indoor peak higher than outdoors: the house overheats.

This house: Living + dining -3.9 °C Intermediate 4/5 | Bedroom 1 -4.0 °C Intermediate 4/5 | Bedroom 2 -4.8 °C Superior 5/5 | Bedroom 3 -4.9 °C Superior 5/5

Source: ABNT NBR 15575-1:2013 (Residential buildings - Performance), daylight and thermal requirements. Applies to bedrooms and living rooms. We use the hottest day of the weather file, not the norm's standard design day.

Coldest day: indoor minimum vs. outdoor minimum (NBR 15575) Brazilian norm

On the coldest day of the typical year, the coldest indoor air temperature compared with the coldest outdoor temperature. Positive means the house is warmer than the outdoors.

Fails 1/5	below 3 degC (indoor min - outdoor min)	Less than 3 degC warmer than outdoors.
Minimum 3/5	3 to 5 degC (indoor min - outdoor min)	At least 3 degC warmer than outdoors.
Intermediate 4/5	5 to 7 degC (indoor min - outdoor min)	At least 5 degC warmer than outdoors.
Superior 5/5	7 or more degC (indoor min - outdoor min)	At least 7 degC warmer than outdoors.

This house: Living + dining +9.4 °C Superior 5/5 | Bedroom 1 +10.3 °C Superior 5/5 | Bedroom 2 +10.7 °C Superior 5/5 | Bedroom 3 +10.8 °C Superior 5/5

Source: ABNT NBR 15575-1:2013 (Residential buildings - Performance), daylight and thermal requirements. Winter check is required in bioclimatic zones 1 to 5, which includes Maringa.

Hottest hour of the year inside Rule of thumb

Highest operative temperature (air + wall surfaces) reached in the room.

Comfortable 5/5	below 28 degC	Never really hot.
Warm 4/5	28 to 30 degC	Warm on the hottest days; a fan is enough.
Hot 3/5	30 to 32 degC	Hot spells; fans help, sleep is affected.
Very hot 2/5	32 to 35 degC	Heat stress risk for children and the elderly.
Extreme 1/5	35 or more degC	Dangerous without cooling.

This house: Living + dining 33.7 °C Very hot 2/5 | Hallway 29.1 °C Warm 4/5 | Bedroom 1 33.6 °C Very hot 2/5 | Bathroom 1 35.5 °C Extreme 1/5 | Bedroom 2 32.8 °C Very hot 2/5 | Bathroom 2 32.5 °C Very hot 2/5 | Bedroom 3 32.7 °C Very hot 2/5 | Kitchen 35.2 °C Extreme 1/5 | Outdoor bath 36.6 °C Extreme 1/5 | Outdoor storage 35.8 °C Extreme 1/5

Source: ASHRAE 55 adaptive comfort model (80% / 90% acceptability); Rule of thumb of this study (no norm gives a number). The adaptive-comfort upper limit in a warm month is about 29 degC (80% acceptability).

Coldest hour of the year inside International standard

Lowest operative temperature reached in the room.

Dangerous 1/5	below 12 degC	Cold stress; risk of heart and breathing problems.
Cold 2/5	12 to 16 degC	Below 16 degC the risk of respiratory problems rises.
Cool 3/5	16 to 18 degC	A jumper is needed; below the WHO 18 degC recommendation.
Good 4/5	18 to 20 degC	Meets the WHO recommendation of at least 18 degC.
Pleasant 5/5	20 or more degC	Never uncomfortably cold.

This house: Living + dining 12.9 °C Cold 2/5 | Hallway 16.5 °C Cool 3/5 | Bedroom 1 13.1 °C Cold 2/5 | Bathroom 1 11.1 °C Dangerous 1/5 | Bedroom 2 14.0 °C Cold 2/5 | Bathroom 2 14.4 °C Cold 2/5 | Bedroom 3 14.0 °C Cold 2/5 | Kitchen 11.5 °C Dangerous 1/5 | Outdoor bath 10.9 °C Dangerous 1/5 | Outdoor storage 10.8 °C Dangerous 1/5

Source: WHO Housing and Health Guidelines 2018 (indoor temperature at least 18 C).

Electricity to keep the room cool with an air conditioner Rule of thumb

Electricity a split air conditioner would use in a year to hold the room at 26 C (upper limit of the summer comfort zone of NBR 16401-2) during the hours the room is used (bedrooms 21:00-08:00, living 14:00-23:00, kitchen 08:00-21:00). A room that overheats in summer costs more here. Cost = kWh x tariff.

Very low 5/5	below 10 kWh per m2 per year	Rarely needs cooling.
Low 4/5	10 to 25 kWh per m2 per year	A few weeks of cooling a year.
Moderate 3/5	25 to 45 kWh per m2 per year	Cooling needed through most of the summer.
High 2/5	45 to 70 kWh per m2 per year	Expensive to keep cool: the room overheats a lot.
Very high 1/5	70 or more kWh per m2 per year	Very expensive: shade, insulation or a better roof are worth more than a bigger air conditioner.

This house: Living + dining 18.8 kWh/m^2 Low 4/5 | Bedroom 1 11.0 kWh/m^2 Low 4/5 | Bedroom 2 8.1 kWh/m^2 Very low 5/5 | Bedroom 3 7.9 kWh/m^2 Very low 5/5 | Kitchen 33.8 kWh/m^2 Moderate 3/5

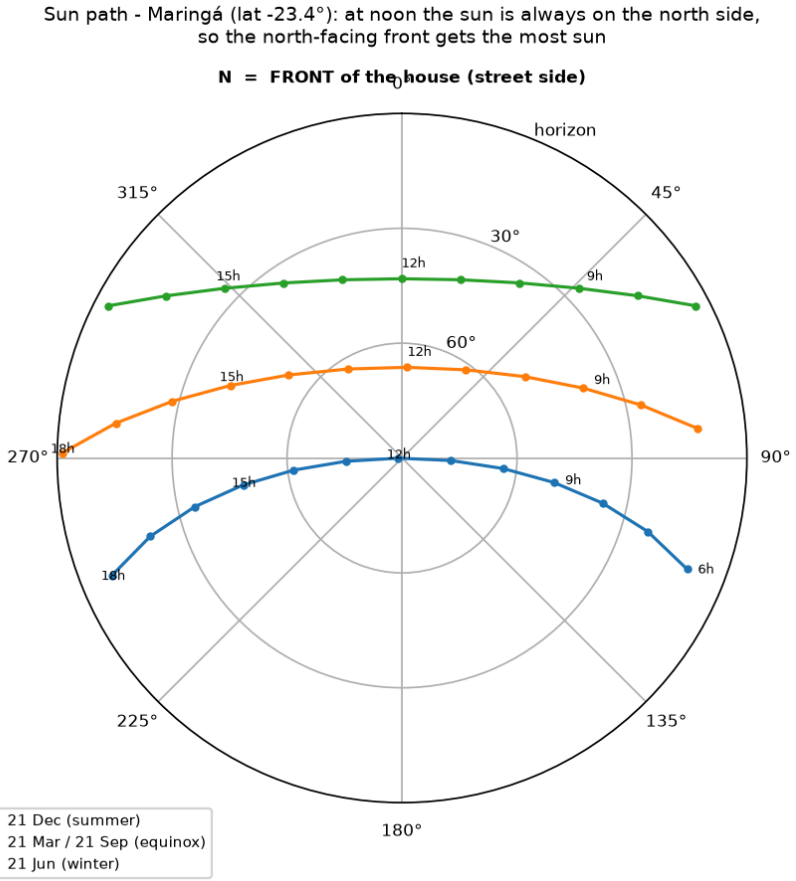
Source: Rule of thumb of this study (no norm gives a number); ABNT NBR 16401-2 (air conditioning, indoor comfort): summer zone about 22.5-25.5 C at 65% RH and 23-26 C at 35% RH. Assumes COP 3.0 (Inmetro class A split of about 2012), windows closed while the air conditioner runs, sensible heat only (no dehumidification) so real bills are somewhat higher. Tariff is an assumption, see the report.

6. Sun on the outside, walls and roof

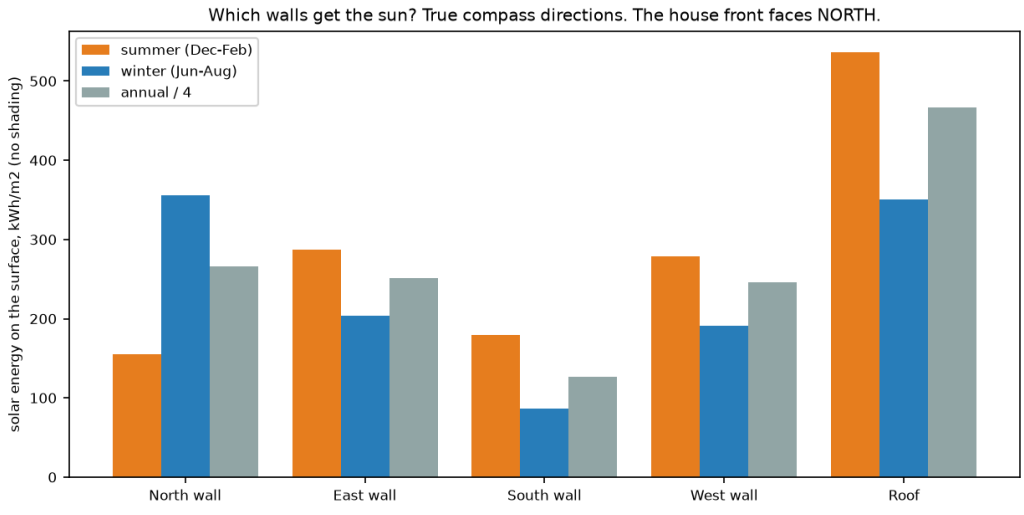
The house faces north: in Maringá the sun is in the north half of the sky all year, so the front gets lots of winter sun (good) and little summer sun, while east and west walls take strong summer sun (bad).

6.1 Sun energy on each surface (kWh per m², no shading)

Surface	Whole year	Summer (Dec-Feb)	Winter (Jun-Aug)
North wall	1066	155 Moderate 4/5	356 High 5/5
East wall	1004	287 High 3/5	204 Moderate 3/5
South wall	507	179 Moderate 4/5	86 Low 2/5
West wall	982	279 High 3/5	191 Moderate 3/5
Roof (horizontal)	1865	536	351



Path of the sun over the year; N is the front (street side).



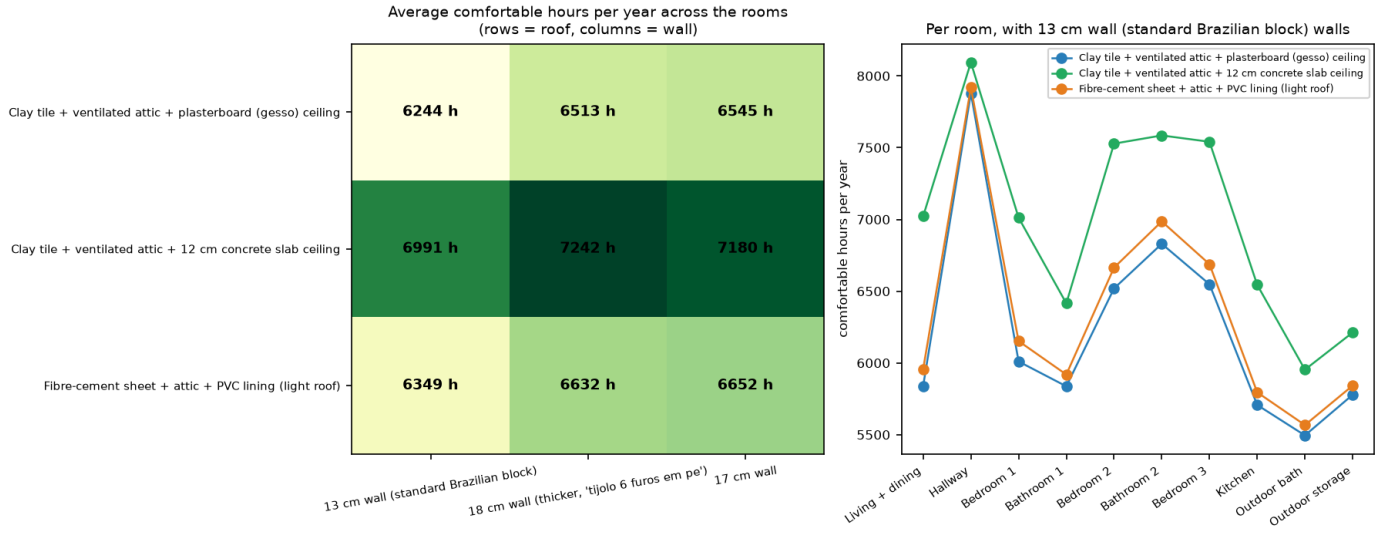
Solar energy on each wall by season.

6.2 Do the walls and roof meet NBR 15575?

Item	This house	Limit (NBR 15575)	Result
External wall heat transmittance U Ceramic hollow brick 8-hole 9x19x19 cm + 2 cm plaster both sides = 13 cm wall (standard Brazilian block) (paint / roof absorptance 0.45)	2.52 W/m²K	<= 3.7	meets
External wall heat capacity Ceramic hollow brick 8-hole 9x19x19 cm + 2 cm plaster both sides = 13 cm wall (standard Brazilian block) (paint / roof absorptance 0.45)	134 kJ/m²K	>= 130	meets
Roof heat transmittance U (indicative) Clay tile + ventilated attic + 12 cm concrete slab ceiling (paint / roof absorptance 0.7)	1.78 W/m²K	<= 1.5	does not meet

6.3 What if the materials change?

Walls	Roof / ceiling	Avg. comfortable hours	Too hot	Too cold	Cooling bill
Ceramic hollow brick 14 cm + 2 cm plaster both sides = 18 cm wall (thicker, 'tijolo 6 furos em pe')	Clay tile + ventilated attic + 12 cm concrete slab ceiling	7,242 h = 83% Good 4/5	747 h	771 h	R$ 1,377/yr
Concrete block 14 cm + 1.5 cm plaster both sides = 17 cm wall	Clay tile + ventilated attic + 12 cm concrete slab ceiling	7,180 h = 82% Good 4/5	805 h	775 h	R$ 1,568/yr
Ceramic hollow brick 8-hole 9x19x19 cm + 2 cm plaster both sides = 13 cm wall (standard Brazilian block)	Clay tile + ventilated attic + 12 cm concrete slab ceiling	6,991 h = 80% Good 4/5	928 h	842 h	R$ 1,600/yr
Concrete block 14 cm + 1.5 cm plaster both sides = 17 cm wall	Fibre-cement sheet + attic + PVC lining (light roof)	6,652 h = 76% Good 4/5	1,112 h	997 h	R$ 1,700/yr
Ceramic hollow brick 14 cm + 2 cm plaster both sides = 18 cm wall (thicker, 'tijolo 6 furos em pe')	Fibre-cement sheet + attic + PVC lining (light roof)	6,632 h = 76% Good 4/5	1,098 h	1,029 h	R$ 1,524/yr
Concrete block 14 cm + 1.5 cm plaster both sides = 17 cm wall	Clay tile + ventilated attic + plasterboard (gesso) ceiling	6,545 h = 75% Acceptable 3/5	1,231 h	985 h	R$ 1,856/yr
Ceramic hollow brick 14 cm + 2 cm plaster both sides = 18 cm wall (thicker, 'tijolo 6 furos em pe')	Clay tile + ventilated attic + plasterboard (gesso) ceiling	6,513 h = 74% Acceptable 3/5	1,230 h	1,017 h	R$ 1,680/yr
Ceramic hollow brick 8-hole 9x19x19 cm + 2 cm plaster both sides = 13 cm wall (standard Brazilian block)	Fibre-cement sheet + attic + PVC lining (light roof)	6,349 h = 72% Acceptable 3/5	1,270 h	1,141 h	R$ 1,800/yr
Ceramic hollow brick 8-hole 9x19x19 cm + 2 cm plaster both sides = 13 cm wall (standard Brazilian block)	Clay tile + ventilated attic + plasterboard (gesso) ceiling	6,244 h = 71% Acceptable 3/5	1,388 h	1,128 h	R$ 1,961/yr



Comfortable hours with different walls and roofs.

What counts as good? (sun and construction)

Summer sun on a wall (Dec-Feb) Rule of thumb

Solar energy on a vertical wall with no shading in December-February. Lower is better: it is heat you do not want.

Low 5/5	below 150 kWh/m2	Little summer heat from the sun.
Moderate 4/5	150 to 225 kWh/m2	Some heat; an eave or a light colour handles it.
High 3/5	225 to 300 kWh/m2	Hot wall: needs shading, light colour, or insulation.
Very high 2/5	300 or more kWh/m2	Very hot wall.

This house: North wall 155 Moderate 4/5 | East wall 287 High 3/5 | South wall 179 Moderate 4/5 | West wall 279 High 3/5

Source: Rule of thumb of this study (no norm gives a number). Roughly 1.7 / 2.5 / 3.3 kWh/m2 per day. The norm handles wall heat through limits on U-value and paint colour.

Winter sun on a wall (Jun-Aug) Rule of thumb

Solar energy on a vertical wall in June-August. Higher is better in a cool winter: free heat (NBR 15220-3 recommends winter solar heating for zone 3).

Low 2/5	below 120 kWh/m2	Almost no winter sun.
Moderate 3/5	120 to 220 kWh/m2	Some winter sun.
Good 4/5	220 to 320 kWh/m2	Good winter sun.
High 5/5	320 or more kWh/m2	A lot of free winter heat.

This house: North wall 356 High 5/5 | East wall 204 Moderate 3/5 | South wall 86 Low 2/5 | West wall 191 Moderate 3/5

Source: Rule of thumb of this study (no norm gives a number); ABNT NBR 15220-3: Brazilian bioclimatic zoning and building guidelines (Maringa is generally listed in zone 3).

Wall heat transmittance U (NBR 15575-4) Brazilian norm

How fast heat flows through the wall (lower = better insulated). Limit in zones 3-8: 3.7 if the outside paint is light (solar absorptance up to 0.6), 2.5 if darker.

Meets both limits 5/5	below 2.5 W/m2K	Meets the limit even with dark paint.
Light paint only 4/5	2.5 to 3.7 W/m2K	Meets the limit only with light-coloured paint (absorptance 0.6 or less).
Fails 1/5	3.7 or more W/m2K	Above the limit: the simplified method is not met.

This house: External wall heat transmittance U 2.52 (limit <= 3.7) Light paint only 4/5

Source: ABNT NBR 15575-4:2021 amendment draft (Mar 2021): simplified method for walls, ventilation, transparent areas.

Wall heat capacity (NBR 15575-4) Brazilian norm

How much heat the wall stores (higher = calmer indoor temperature). Limit: at least 130 in zones 1-7.

Fails 1/5	below 130 kJ/m2K	Less heat storage than the minimum.
Meets 5/5	130 or more kJ/m2K	Meets the minimum of 130.

This house: External wall heat capacity 134 (limit >= 130) Meets 5/5

Source: ABNT NBR 15575-4:2021 amendment draft (Mar 2021): simplified method for walls, ventilation, transparent areas.

Roof heat transmittance U (NBR 15575-5) Brazilian norm

Heat flow through roof + ceiling. Limit in zones 3-6: 2.3 if the roof is light-coloured (absorptance up to 0.6), 1.5 if darker (e.g. red clay tile).

Meets both limits 5/5	below 1.5 W/m2K	Meets the limit even for a dark roof.
Light roof only 4/5	1.5 to 2.3 W/m2K	Meets the limit only if the roof surface is light.
Fails 1/5	2.3 or more W/m2K	Above the limit.

This house: Roof heat transmittance U (indicative) 1.78 (limit <= 1.5) Light roof only 4/5

Source: ABNT NBR 15575-5:2021 amendment draft: simplified method for roofs. Our roof U is indicative: a ventilated attic is not modelled by the standard calculation.

7. Assumptions, limits and sources

- **Dimensions** come from a video scan plus the listing; window sizes are common catalogue sizes of about 2012, not measured. This is not a survey.
- **Materials** are the standard Brazilian 8-hole ceramic block with plaster and a clay-tile roof with concrete slab ceiling, not confirmed for this house. Neighbouring buildings, street trees and curtains are not modelled.
- **Doors** are assumed open for light and air. Furniture, people, appliances and lighting heat are not included (empty house).
- The comfort model is a simplified hourly heat balance, not calibrated against measurements: use it to compare rooms and options. EnergyPlus would be the next level of rigour.
- **Wind** data are for open terrain at 10 m; inside a plot with walls and neighbours the speeds are lower. No CFD was run.
- **Daylight** uses an isotropic sky; real partly-cloudy skies are brighter near the sun.
- Rating scales marked *Rule of thumb* are ours; the ones marked norm come from the documents listed below. Several norm values were read from public summaries and amendment drafts: confirm them in the full ABNT texts before using them in a legal, contractual or financing decision.

Sources

- ABNT NBR 15575-1:2013 (Residential buildings - Performance), daylight and thermal requirements
- ABNT NBR 15575-1:2021 + amendment (thermal performance by simulation vs. a reference house)
- ABNT NBR 15575-4:2021 amendment draft (Mar 2021): simplified method for walls, ventilation, transparent areas
- ABNT NBR 15575-5:2021 amendment draft: simplified method for roofs
- ABNT NBR 15220-3: Brazilian bioclimatic zoning and building guidelines (Maringa is generally listed in zone 3)
- ABNT NBR 5413:1992 (withdrawn in 2013 but still the usual Brazilian reference for HOME lighting levels)
- ABNT NBR ISO/CIE 8995-1:2013 (workplace lighting; reading/writing tasks 300-500 lx)
- EN 17037:2018 Daylight in buildings (UK National Annex dwelling targets 100/150/200 lx; sunlight 1.5/3/4 h)
- IES LM-83-12 (sDA and ASE), as used by LEED v4.1
- ASHRAE 55 adaptive comfort model (80% / 90% acceptability)
- ABNT NBR 16401-2 (air conditioning, indoor comfort): summer zone about 22.5-25.5 C at 65% RH and 23-26 C at 35% RH
- WHO Housing and Health Guidelines 2018 (indoor temperature at least 18 C)
- WMO Beaufort wind scale
- Rule of thumb of this study (no norm gives a number)

Run it again after changing the model

Edit the 3D model (house.ifc), then double-click **run_all.bat** in the House3D folder. It redoes the simulation, this PDF and the data for the 3D viewer (viewer / open_viewer.bat).